An Interpretable SAR Image Filtering Algorithm

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***Abstract*— Effective noise suppression is crucial for the subsequent interpretation tasks of synthetic aperture radar (SAR) imagery. Traditional SAR image processing techniques often overlook the coherent nature of noise, leading to a loss of vital detail during filtering. With advancements in deep learning (DL), significant strides have been made in image processing. However, existing DL methods do not fully leverage the imaging mechanisms of SAR, resulting in a lack of specificity and interpretability in the filtering process. To balance noise reduction with detailed preservation and to address the "black box" issue in filtering, we propose an interpretable filtering method that employs a correlation-based upward search for density peaks. Initially, we develop a MeanShift-Markov random field (MS-MRF) filter that integrates MeanShift with MRF in the joint spatial–spectral domain, ensuring both correlation and detailed preservation; the derivation of the MS-MRF filter is rigorously grounded in mathematical theory. Subsequently, we integrate MS-MRF with convolutional operations in DL to create a novel convolutional filter, interpretable MS-MRF convolution (IMMC), which enhances the model's interpretability, noise reduction capabilities, and detailed retention. Extensive experiments demonstrate that our method outperforms state-of-the-art (SOTA) SAR denoising techniques, achieving an average structural similarity index (SSIM) of over 85.00% and an average peak signal-to-noise ratio (PSNR) exceeding 35.00 dB across synthetic datasets with varying noise levels, showing significant improvements in noise suppression, detailed preservation, and interpretability.**

***Index Terms*— Deep learning (DL), interpretability, speckle, synthetic aperture radar (SAR) filter.**

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I. Introduction

SYNTHETIC aperture radar (SAR) can achieve all-weather imaging and data collection capabilities, providing high-resolution images that facilitate precise terrain retrieval [1]. Therefore, they are widely used in target detection, disaster monitoring, and ground exploration [2], among other fields. The coherence of SAR systems tends to produce coherent speckle in images [3]. Speckle in SAR images manifests as alternating light and dark spots, which can cover detailed information in the image, reducing visual effects and affecting the accuracy of tasks based on SAR images such as interpretation, target detection, change detection, and classification. Therefore, it is crucial to suppress noise while preserving texture and edge details. Since the 1970s, after 50 years of development of filtering algorithms, SAR images can be divided into two major categories based on noise reduction characteristics: traditional noise reduction algorithms and deep-learning (DL)-based noise reduction algorithms. Traditional noise reduction methods are based on the statistical relationship between noise and clean signals, and traditional filtering algorithms include spatial-domain filtering algorithms and transform-domain filtering algorithms [3], [4].

SAR image spatial filtering algorithms are divided into local and nonlocal (NL) filters based on the range of pixel neighborhoods. Local filtering methods, such as Lee filters [5], Frost filters [6], and Kuan filters [7], [8], [9], assume that the signal within a window centered on the target pixel is stationary, adjusting the estimation window based on local image statistics to handle signal instability. Although these filters perform well in homogeneous pixel areas (areas with similar backscattering mechanisms), they perform poorly in heterogeneous areas (areas with large differences in backscattering similarity). SAR image local filters may lose image edge detailed information, leading to smooth edges and blurred features.

Transform-domain-based image denoising, such as wavelet transform [10], curvelet transform [11], and signal transform algorithms [12], improves the retention of edge and detailed features. Despite their remarkable efficacy in preserving specific details, these methods are confronted with challenges in establishing accurate a priori estimations for diverse regions and sensor targets.

NL filters [13], [14], [15], [16], [17], [18], [19] select similar pixels/color blocks based on similarity criteria and collect and combine them to reduce speckle. However, these filters have

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high computational costs. Touzi [20] and Deledalle et al. [21] have provided a systematic review of the aforementioned spatial filters.

With the vigorous development of artificial intelligence in the field of computer vision, researchers have begun to focus on how to achieve efficient SAR image filtering using DL [22] and have also achieved significant results, such as SAR-CNN [23], SIFSDNET [24], Sublook2Sublook [25], PCENet [26], and CL-SAR [27]. These methods can automatically learn hidden nonlinear relationships without relying on handcrafted parameters.

Despite the great success of DL-based SAR denoising, there are still some issues that limit its development, such as the "black box effect." The lack of mathematical interpretation of the internal mechanism of the model and the opaque working process have become a bottleneck in the application of SAR images in downstream tasks. To fundamentally enhance the applicability of SAR images, it is necessary to overcome the "black box effect" of deep neural networks (DNNs).

In the realm of traditional SAR image processing, both conventional algorithm-based filtering and DL-based filtering often fail to adequately consider the unique properties of speckle noise inherent in SAR imagery. This results in a challenging balance between noise suppression and detailed preservation within the spatial domain. Although existing DL-based SAR denoising methods have significantly outperformed traditional filtering methods in terms of quality, they lack a physically meaningful mathematical interpretation during the filtering process, which limits a deeper understanding of the image content and its effectiveness in subsequent applications. Moreover, current DL methods face challenges in optimization and application portability. Denoising based on DL varies depending on noise training and processing levels, and can function both as a standalone denoising algorithm and as an integrated filter guidance with traditional methods. It has been confirmed that the latter is a fast and advanced fusion denoising method, such as SAR-RDCP [28] and MTCSAR [70]. Building on these filtering concepts, this article proposes an interpretable filtering method that combines the advantages of traditional filtering and DL, aiming to balance the performance between noise suppression and detail retention, while enhancing the interpretability of the filtering process.

The filtering method proposed in this article initially constructs a MeanShift-Markov random field (MS-MRF) filter with correlation and detailed preservation capabilities. This filter is achieved by combining MeanShift [29] and MRF [30] in a joint spatial–spectral domain. The computational process of this filter strictly adheres to mathematical theory and, by integrating with convolutional operations in DL, constructs a new convolutional filter, interpretable MS-MRF convolution (IMMC), to enhance the model's interpretability and denoising effect. Additionally, this method strengthens the correlation between pixels and the model's ability to preserve details. By replacing the 3×3 convolution in the original ResNet bottleneck with IMMC, a new residual filtering block, interpretable MS-MRF block (IMMB), is formed, which inherits the advantages of the original residual block in being easy to optimize and portable to other networks. The contributions of this article include the following.

1) From the Perspective of Denoising: Addressing the critical challenge of achieving a balance between speckle noise mitigation and image quality preservation in SAR image filtering, this article innovatively proposes an MS-MRF filter. By integrating MeanShift with MRF, the filter employs an upward search strategy predicated on the peak of correlation density, segmenting the image into multiple regions of varying dimensions. This method has strong adaptability and can more robustly filter the homogeneous and heterogeneous regions in SAR images. It has a correlation partitioning filter for the texture and edge contour of different regions, thus alleviating the problem of texture blur and excessive smoothing, and improving the fidelity of the image.

In response to the challenge of balancing noise suppression with image quality preservation in traditional SAR image filtering, this article introduces a novel MS-MRF filter that integrates MeanShift with MRF. This filter operates in the spatial–spectral domain, effectively combining local and global information, thereby overcoming the limitations of traditional filters that are constrained by local spatial correlations. This innovative approach not only enhances filtering performance, but also improves the detailed expressiveness of images, marking a new technological breakthrough in the field of SAR image processing.

2) From the Perspective of Interpretability: In the field of SAR image processing, preserving detailed information has always been a challenge. Addressing this challenge, the MS-MRF filter, constructed in the joint space–spectral domain, ingeniously integrates the local sensitivity of the MeanShift algorithm with the global information processing capability of the MRF algorithm. This integration effectively overcomes the limitations of traditional filters, which rely excessively on local spatial correlation, and also demonstrates a certain effect in enhancing image details.

The proposed residual filter block, IMMB, inherits the advantages of the original ResNet residual block, such as ease of optimization and portability. Incorporating this module into the training regimen of DL frameworks facilitates the elucidation of the filtering mechanism.

II. Related Works

Interpretability is a relatively concrete and vivid concept, referring to the ability of humans to perceive the attributes and operating principles of a model through data input analysis, based on structure and parameters [31]. Essentially, interpretability is a bridge between artificial intelligence and human wisdom. Explaining the complex reasoning of DNN from different perspectives will help DNN simulate the human brain.

Traditional machine learning models simulate intelligent behavior patterns in the real physical world based on mathematical logic established on a set of assumptions and statistical rules [32]. Mathematical models are interpretable but have limited ability to model high-dimensional data. The gap between them and physical models determines their performance and reliability [33]. DNNs based on end-to-end connections have strong data-fitting capabilities. However, their complex and

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redundant structures and opaque operational rules make them lack interpretability [34].

The effectiveness of DL models mainly depends on the quality of data and its effective representation [22]. Tedious raw data need to be transformed into a form that can be processed by the model to accelerate convergence [35]. Therefore, the MS-MRF correlation filter proposed in this article has considerable application value.

Molnar [36] systematically elaborated on the criteria for model interpretability in "interpretable machine learning." Interpretability can be divided into two dimensions according to model design characteristics and explanation scale.

1) According to whether the model's interpretability is built-in or additional, interpretability can be divided into two modes: inherent and posthoc. Inherent interpretability refers to the model structure itself being easy to understand, such as data-driven represented by mathematical logic (linear models, parameter models, or tree-based models [37]), which allows people to intuitively understand the decision-making process and foundation. In contrast, posthoc interpretability involves applying interpretive methods to already selected and trained black box models (such as ensemble methods or neural networks), such as activation maps, to increase the model's interpretability.
2) According to the scale of explanation, interpretability is divided into local and global scales. Global interpretability: Global explanations help us understand the entire conditional distribution modeled by the trained response function, but global explanations can be approximate or based on averages. Local explainability: Local explanations promote understanding of a single data point or a small range of distributions, and this classification of explanations discusses whether the interpretation method explains a single prediction or the entire model behavior.

In the realm of DNNs, certain methodologies have been developed to ascribe specific meanings to neurons or convolutional layers, thereby integrating mathematical logic or physical interpretability into DNNs, as discussed in [38]. This approach yields a hybrid form of interpretability. Sparse, physics-based, and partially interpretable neural networks (SPINNs) utilize physical equations as operational constraints for solving partial differential equations (PDEs), as detailed in [39]. Interpretability is transformed into the visualization of weights and biases, elucidating the overall architecture and specific parameters. Neural additive models (NAMs) construct generalized additive models through linear combinations within DNNs, offering enhanced interpretability compared to shallow decision trees and logistic regression, as described in [40]. Similarly, partition of unity networks can learn independent polynomial functions at various nodes, as indicated in [41]. Terpretable nEuraL cLustering (TELL) achieves discrete parallel computation of k-means through differentiable neural networks, as referenced in [42]. These efforts begin with the structure of DNNs, endowing nodes with mathematical and physical significance to interpret the weights and biases within hyperparameters.

In the field of speckle removal for SAR images, DL models based on hybrid interpretability have increasingly garnered the attention of researchers. In 2024, the model SAR-DURNet [43], which combines DL with traditional methods, leveraged prior knowledge of noise distribution to formulate an optimization problem for speckle noise removal and was end-to-end trained using a simulated SAR image dataset. SAR-DURNet effectively removes speckle noise and learns complex patterns from noisy images using DL techniques, but its decision-making process lacks intuitive interpretability. Trans-NLM network [44] is a rare interpretable SAR filtering model that integrates Transformer architecture into traditional denoising algorithms based on NLM filtering. This integration enhances the despeckling performance of SAR images while maintaining the interpretability of the algorithm, Trans-NLM performs excellently in speckle enhancement and efficiency, and it also has a more interpretable reasoning process and transfer learning ability. However, as a Transformer-based method, it has a large computational load, which is difficult to optimize and is rarely applicable to other models. The main filtering explanation function of our proposed interpretable SAR filtering model is focused on the SAR filtering bottleneck residual block (IMMB), and the improved interpretable residual filtering IMMB inherits the original residual block's advantages of being easy to optimize and portable.

Summarizing the findings from [37], [39], [66], and [67], the interpretable SAR image model proposed in this article can be categorized as a hybrid of intrinsic and posthoc local interpretability filtering models. That is, it possesses a mathematical interpretation of the filtering process, and numerous experiments have demonstrated the interpretability of neural networks within the activation space [45], [71]. After training, activation maps can be utilized to observe changes in features, thereby gaining a deeper understanding of the interplay among features.

III. Methodology

This section elaborates on the proposed MS-MRF, IMMC, IMMB, and explainability. Fig. 1 illustrates the overall framework of the working principle of IMMB. By combining the locally focused MeanShift and the globally focused MRF, an upward searching density peak correlation MS-MRF filter is obtained, which is detailed in Section III-C; endowing the 3×3 convolution kernel with the mathematical interpretation of MS-MRF results in an inherently explainable IMMC convolution kernel, detailed in Section III-D; replacing the original 3×3 convolution in the first stage of the ResNet with IMMC endows it with inherent explainability, and visualizing the activated model posttraining with activation functions provides it with posthoc explainability, ultimately obtaining a mixed explainability SAR image filter, detailed in Sections III-E and III-F.

A. MeanShift

Given an n-independent and equally distributed random vector $x_i, i = 1, \ldots, n$ in d-dimensional space $\mathbb{R}^d$, a kernel

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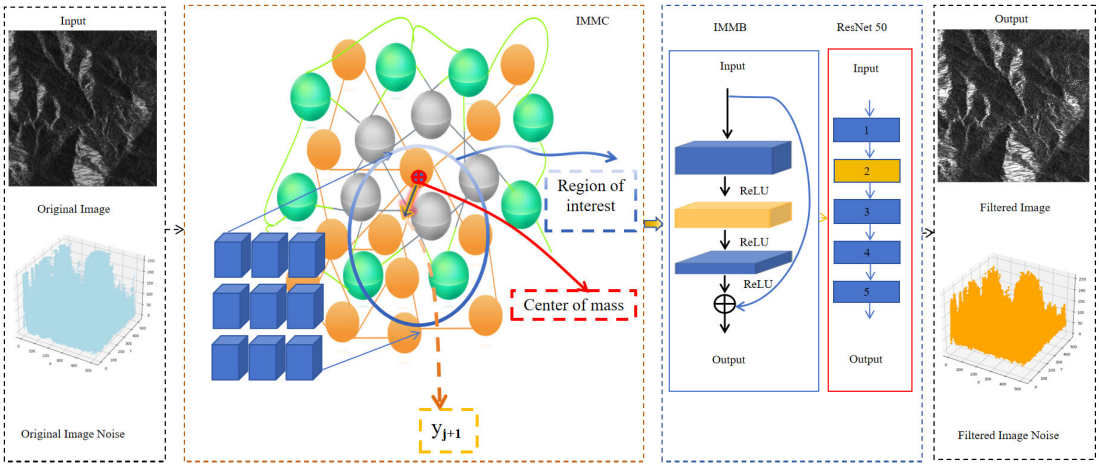


Fig. 1. Overall framework diagram of the working principle of the interpretable filter. Schematic of the working principle of the IMMB. The process within the yellow dashed frame represents the formation of the IMMC; the 3×3 small blue blocks represent the 3×3 convolution kernel, the gray spheres represent pixels within the coarse resolution (CR) pixel matrix, the orange spheres represent pixels within the fine resolution (FR) pixel matrix, and the red circle with a plus sign indicates the center point within the filtering area. The arrows indicate a series of MeanShift vectors pointing to the location y_{j+1}. Vectors pointing to a high-density group. The blue dashed frame indicates the formation process of the IMMB. In the ResNet50 architecture, we have replaced the original first block with IMMB only in the first stage, while the remaining bottleneck blocks in other stages remain unchanged. In the IMMB bottleneck block, the first blue rectangle represents a 1×1 convolution kernel, the second yellow rectangle represents the IMMC that replaces the original 3×3 convolution kernel, and the third blue rectangle represents a 1×1 convolution kernel. In a regular bottleneck block, the first blue rectangle represents a 1×1 convolution kernel, the second blue rectangle represents a 3×3 convolution kernel, and the third blue rectangle represents a 1×1 convolution kernel.

density estimator can be defined as follows:

$$\hat{f}(x) = \frac{1}{n} \sum_{i=1}^{n} K_H(x - x_i) \tag{1}$$

where H represents a $d \times d$-dimensional symmetric positive definite bandwidth matrix, and $K(x)$ is the kernel function. When defining H as $H = h^2 I$. Then, (1) is redefined using $H = h^2 I$, where h is the bandwidth parameter, as shown in the following equation:

$$\hat{f}(x) = \frac{1}{nh^2} \sum_{i=1}^{n} K\left(\frac{x - x_i}{h}\right). \tag{2}$$

The radial symmetric kernel function is defined as shown in the following equation:

$$K(x) = c_{k,d} k(\|x\|^2). \tag{3}$$

In the given context, $c_{k,d}$ represents the normalization constant. The function $k(x)$ is referred to as the profile function of the kernel function $K(x)$, with the condition that $x \geq 0$.

Substituting (3) into (2) yields the following equation:

$$\hat{f}_{h,K}(x) = \frac{c_{k,d}}{nh^2} \sum_{i=1}^{n} K\left(\left\|\frac{x - x_i}{h}\right\|^2\right). \tag{4}$$

The gradient of (4) is calculated, and the density gradient estimator is obtained, as shown in the following equation:

$$\hat{\nabla} f_{h,k}(x) \equiv \hat{\nabla} f_{h,k}(x) = \frac{2c_{k,d}}{nh^{d+2}} \sum_{i=1}^{n} (x - x_i) k'\left(\left\|\frac{x - x_i}{h}\right\|^2\right). \tag{5}$$

The redefined function $q(x)$ is shown in the following equation, and $k'(x)$ is the derivative of $k(x)$, representing the rate of change of $k(x)$:

$$q(x) = -k'(x). \tag{6}$$

Assume that the reciprocal of the profile function $k(x)$ exists for all $x \in [0, \infty)$ except for a finite set of points. $q(x)$ is now taken as a profile function, and its corresponding function can be expressed as follows:

$$Q(x) = c_{q,d} q\left(\|x\|^2\right). \tag{7}$$

In the given context, $c_{q,d}$ represents the normalization constant. The kernel function $K(x)$ is referred to as the shadow of $Q(x)$. The role of the shadow function $K(x)$ is to update the mean shift vector in the MeanShift algorithm. In the MeanShift algorithm, we seek the point that maximizes the kernel density estimation, a process that involves adjusting the center of the search window to move toward areas of higher density. To achieve this, we calculate the gradient of the density and then use this gradient to update the center of the search window.

Now, substitute (6) into (5) to obtain the following equation:

$$\begin{aligned} &\nabla \hat{f}_{h,k}(x) \\ &= \frac{2c_{k,d}}{nh^{d+2}} \sum_{i=1}^{n} (x_i - x) q\left(\left\|\frac{x - x_i}{h}\right\|^2\right) \\ &= \frac{2c_{k,d}}{nh^{d+2}} \left[\sum_{i=1}^{n} q\left(\left\|\frac{x - x_i}{h}\right\|^2\right)\right] \left[\frac{\sum_{i=1}^{n} x_i q\left(\left\|\frac{x - x_i}{h}\right\|^2\right)}{\sum_{i=1}^{n} q\left(\left\|\frac{x - x_i}{h}\right\|^2\right)} - x\right]. \end{aligned} \tag{8}$$

The formula in the previous version of (8) represents the density estimate as follows:

$$\hat{f}_{h,Q}(x) = \frac{2c_{k,d}}{nh^{d+2}} \sum_{i=1}^{n} q\left(\left\|\frac{x - x_i}{h}\right\|^2\right). \tag{9}$$

The second half of (8) is divided into the MeanShift, which represents the difference between the weighted average

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$m_{h,Q}(x)$ weighted by the kernel function $Q(x)$ in the kernel function window and the window center x as shown in the following equation:

$$M_{h,Q}(x) = \frac{\sum_{i=1}^{n} x_i q\left(\left\|\frac{x-x_i}{h}\right\|^2\right)}{\sum_{i=1}^{n} q\left(\left\|\frac{x-x_i}{h}\right\|^2\right)} - x = m_{h,Q}(x) - x. \quad (10)$$

Substituting (9) and (10) into (8) yields (11) as follows:

$$\nabla \hat{f}_{h,K}(x) = \hat{f}_{h,Q}(x)\frac{2c_{k,d}}{h^2 c_{k,d}} M_{h,Q}(x) \quad (11)$$

$$M_{h,Q}(x) = \frac{1}{2}h^2 c \frac{\nabla \hat{f}_{h,K}(x)}{\hat{f}_{h,Q}(x)}. \quad (12)$$

When applying the MeanShift filter to SAR images, selecting the intensity (or magnitude) of the scattering vector as the input for MeanShift leads to the issue of how to set the bandwidth value in the spectral domain. Since the spectral information of SAR is different from that of optical imagery, it is best to abandon the inherent thinking about optical images. In optical images, the bandwidth is typically stored as integer values within the range $[0, 255]$ due to its small storage size, and a single bandwidth is sufficient for MeanShift image processing. However, the intensity of SAR images is often stored as floating-point data with values that are not fixed, theoretically ranging from $[0, \infty)$. Therefore, we adopt an adaptive variable bandwidth.

Assuming $\mathbf{x}_i$ is the scattering vector of a p-dimensional original polarimetric SAR intensity (or amplitude) image, a series of d-dimensional vectors $\mathbf{x}_i$ (where $d = p + 2$) defined in the spatial–spectral joint domain can be constructed. Each vector corresponds to a pixel, with the 2-D spatial information existing in the spatial domain, represented by x_s as the spatial-domain part of the vector x, and the spectral information existing in the spectral domain, represented by x_r as the spectral-domain part of the vector.

Because we have proposed an adaptive bandwidth, different bandwidths are adopted for each regional center point x, that is, $h = h(x_i)$, thereby obtaining the Balloon density estimator

$$\nabla \hat{f}_B(x) = \frac{1}{nh(x)^d} \sum_{i=1}^{n} k\left(\frac{x - x_i}{h(x_i)}\right). \quad (13)$$

As we have previously discussed the nonfixed storage of SAR data and the use of adaptive bandwidth, we discuss the bandwidth categorization as follows:

$$h_i = \begin{cases} h_1, & \text{if } \mathbf{x}_{i,j} \le x_j \\ h_2, & \text{if } \mathbf{x}_{i,j} > x_j, \end{cases} \quad 1 \le j \le p. \quad (14)$$

The notation $x_{i,j}$ represents the value of the ith pixel at the jth spectral channel. $h_{j,1}$ and $h_{j,2}$ denote the bandwidth values corresponding to the jth spectral channel when $x_{i,j} \le x_j$ and $x_{i,j} > x_j$, respectively, where p indicates the dimensionality of the spectral domain

$$M_{\boldsymbol{h}_s,\boldsymbol{h}_r,Q}(x) = \frac{\sum_{i=1}^{n} q_s\left(\left\|\frac{x_{i,s}-x_s}{h_s}\right\|^2\right) q_r\left(\left\|\frac{x_{i,r}-x_r}{h_r}\right\|^2\right) x_i}{\sum_{i=1}^{n} q_s\left(\left\|\frac{x_{i,s}-x_s}{h_s}\right\|^2\right) q_r\left(\left\|\frac{x_{i,r}-x_r}{h_r}\right\|^2\right)} - x$$
$$= m_{h_s,h_r,Q}(x) - x. \quad (15)$$

B. MRF

The MRF [46] is a random field with Markov characteristics, and the value of each node is related only to the node of the collar region, which is used to model the probabilistic model of the relationship between random variables. MRF treats the image as a whole by defining interactions between pixels, where the value of each pixel depends not only on its own characteristics but is also influenced by the values of surrounding pixels. The advantage of MRF in image denoising lies in its ability to consider both local features (such as edges and textures) and global features (such as regional consistency) simultaneously, thereby preserving important structural information of the image while removing noise.

C. MS-MRF

The MS-MRF combines the MeanShift and the MRF to form a filter. The MeanShift algorithm is commonly used for image segmentation, while the MRF is a probabilistic graph model that is often used for tasks such as image denoising and image restoration. By combining them, the spatial clustering property of the MeanShift and the probabilistic modeling property of the MRF can be utilized to achieve a more efficient filtering process.

To realize the MS-MRF filter, the MeanShift algorithm is first used for spatial clustering of images, the MRF algorithm is then used for probabilistic modeling of each pixel, and finally, the MS-MRF filter is generated by combining the results of both, as shown in Fig. 2.

The multivariate kernel function defined in the spatial–spectral joint domain is introduced into the MeanShift algorithm; that is, an image containing p channels can be regarded as a 2-D grid composed of p-dimensional vectors, where each appropriate amount corresponds to a pixel. The position of the pixel, that is, the spatial information, exists in the spatial domain, and the value of the pixel, that is, the spectral information, exists in the spectral domain. If both the spatial domain and the spectral domain use Euclidean metrics, to use both spatial and spectral information, both need to be normalized to synthesize $d = p + 2$.

Markov and conditional random fields (CRFs), i.e., the concept of virtual features, are introduced in the $d = p + 2$ dimensional spatial–spectral joint domain. Suppose the same SAR image is composed of a finer resolution pixel array and a coarser resolution pixel array. Let $I \subset d$ be the pixel cell at FR and $\mathcal{S} \subset d$ be the pixel cell at CR. CR contains ρ^2 times the number of FR pixels, as shown in the following equation, which is the linear relationship of I where S and P represent the linear resolution ratio between the two grids:

$$\frac{S}{I} = \rho^2. \quad (16)$$

Assuming that each pixel in the spatial–spectral joint domain is independently and equally distributed, the relationships between the pixels and cells are as follows: $x_i \in I$ represents a random vector x_i defined in the joint spatial–spectral domain belonging to the fine resolution I, $x_s \in S$ indicates that the spatial vector x_s defined in the

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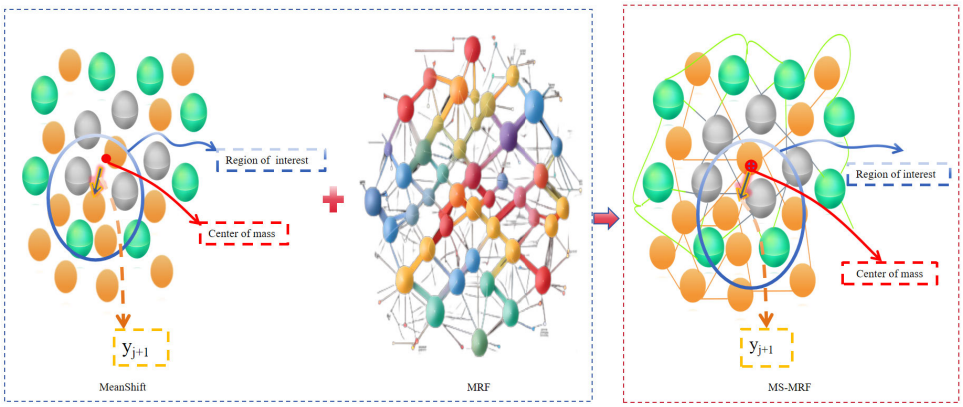


Fig. 2. Formation principle of the MS-MRF.

spatial–spectral joint domain belongs to the coarse resolution S, and $x_r \in S$ indicates that the spectral vector x_r defined in the spatial–spectral joint domain belongs to the coarse resolution S.

In the context of image analysis, latent variables typically refer to pixel features that are not directly observable from the imagery but can exert a significant influence on the outcome of filtering processes. It is postulated that v_i represents a latent variable within SAR image, with $v_i \in I$, denoting that v_i is defined within the spatial–spectral joint domain and belongs to the fine resolution set I. Furthermore, the logarithmic ratio of v_i to x_i is assumed to be independently and identically distributed. For ease of understanding, we have summarized the above relationships in Algorithm 1.

The conditional probability $p(v_i|x_i)$ of the latent variable v_i given the standard variable x_i can be defined by the following mathematical expression:

$$p(v_i|x_i) = \frac{1}{\sqrt{2\pi\sigma_{\log}^2}} \exp\left(-\frac{\left(\log\left(\frac{v_i}{x_i}\right) - \mu_{\log}\right)^2}{2\sigma_{\log}^2}\right). \quad (17)$$

In this expression, the mean $\mu_{\log}$ is the expected value of $\log(v_i/x_i)$, and $\sigma_{\log}^2$ is the variance of this logarithmic ratio. This normal distribution is used to describe the relationship between the latent variable v_i and the standard variable x_i.

Algorithm 1 Divide Coarse and Fine Resolution Pixels and Their Owning Relationships

Definition:

Given space - spectral joint domain:
$d = p + 2$, $d \subset R$.

DefinitionT:

$i \prec I \in FR, s \prec S \in CR, I \subset R, S \subset R, I + S = d,$
$S \subset I, s \prec S, i \prec I, s \prec I, x_s, x_r \in S.$

Computation:

$x_i \to v_i, v_{i,} \in I,$
$x_s, x_r = \frac{1}{\rho^2} \Sigma i \prec x_i,$
$x_s, x_r = \frac{1}{\rho^2} \Sigma i \prec v_i.$

When considering the random field defined on the finer pixel cell I, denoted by y_i for $i \in \mathbb{Z}$, and the set $Y = \{y_i\}$ for $i \in I$, it represents the varying pixels on the finer pixel cell I. By defining a new CRF model, the relationship between pixels in the spatial–spectral joint domain is articulated within the Bayesian probability graph framework.

The maximum a posteriori (MAP) decision rule is encapsulated as shown in the minimization of an appropriate energy function U, as shown in (18). The energy function U consists of the following three terms: D' denotes the unary potential (pixel-level contribution), which correlates with the class-conditional distribution of the ratio of pixel intensities on a finer grid. This segment typically pertains to local image characteristics, such as the statistical information of pixel values, as well as the alignment between these features and the class-conditional probability distributions. D denotes the unary term associated with latent features, which may pertain to attributes that are not directly observable but can be inferred by the model. These features are often utilized to augment the model's capacity for expression. The final component is the contribution of local neighboring pixels, where this part of the energy function emphasizes the interaction between a pixel and its surrounding pixels, typically used to maintain the smoothness of the image or edge information, thereby preserving the structural characteristics of the image

$$U(Y|I, S) = \beta_1 \sum_{i \in I} D'(x_i|y_i) + \beta_2 \sum_{i \in I} D(v_i|y_i) + \beta_3 \sum_{i,j \in I} W(y_i, y_j|x_i, x_j). \quad (18)$$

In the given context, I represents the set of pixels, and Y represents the set of labels. β_1, β_2, and β_3 are the balancing weight parameters that contribute to various terms. $D'(x_i|y_i)$ is the unary potential related to the pixel value x_i, and $D(v_i \mid y_i)$ represents a unary potential related to the latent vector and the subsequent pointing random vector. y_i and y_j are the labels of the adjacent pixels i and j. The symbols y_j, $j = 1, 2, \ldots, m$, represent a series of MeanShift vectors pointing to points, and y_i represents a pixel in the pointing process of a certain region. Additionally, $W(y_i, y_j|x_i, x_j)$ is the smoothness term, typically related to the interaction potential in MRF. For the smoothness, we use the contrast-sensitive Potts model as shown in the following equation:

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$$W\left(y_i, y_j \middle| x_i, x_j\right) = W'\left(y_i, y_j\right) \exp\left[-\frac{\left\|x_i - x_j\right\|^2}{2h^2}\right]. \quad (19)$$

In the given context, $W'(y_i, y_j) = 1 - \delta(y_i, y_j)$, where δ denotes the Kronecker delta. The bandwidth of the Gaussian kernel is set to the average of the squared distance $\left\|x_i - x_j\right\|^2$. If we let $h \to +\infty$, then $W(y_i, y_j) = 1 - \delta(y_i, y_j)$ and (19) represents an MRF algorithm. In this way, the model not only encourages spatially adjacent pixels to have the same label but also encourages these pixels to be close in the feature space, thereby enhancing the contrast sensitivity of the segmentation.

For the pixel class-conditional distribution of $i \in I$, the generalized Gaussian (GG) model has been selected. If a random variable z is distributed according to the GG distribution, then its probability density function (pdf) is shown in the following equation:

$$g(z \mid \kappa) = \frac{\lambda \Gamma(\frac{3}{\lambda})^{\frac{1}{2}}}{2\sqrt{\kappa_2}\Gamma(\frac{1}{\lambda})^{\frac{3}{2}}} \exp\left[-\frac{\Gamma(\frac{3}{\lambda})^{\frac{\lambda}{2}}}{\Gamma(\frac{1}{\lambda})^{\frac{\lambda}{2}}}\left|\frac{z-\kappa_1}{\sqrt{\kappa_2}}\right|^{\lambda}\right]. \quad (20)$$

In the given context, $\kappa = [\kappa_1, \kappa_2,]$ represents the parameters of the distribution. The shape parameter λ ($\lambda > 0$) is determined univocally by κ_1 and κ_2. The symbol $\Gamma(\cdot)$ is the gamma function. We employ the GG distribution to represent $s \prec i, s \prec I$ and estimate the parameters through the cumulative distribution function during each iteration. The unary potential related to the pixel value x_i, denoted as $D'(x_i|y_i)$, is shown in the following equation:

$$D'_{x_i \in I}(x_i|y_i) = \sum_{i \in I}(y_i|x_i)\rho^2 \log(y_i|\kappa(x_i)). \quad (21)$$

Similarly, D'_{x_s} and D'_{x_r} can be obtained by using the GG distribution in $s \prec i, s \prec I$, as shown in the following equations, respectively, and $x_s, x_r = \rho^2 \sum i \prec x_i$ is defined in Algorithm 1:

$$D'_{x_s \in I}(y_{i,s}|x_{i,s}) = \sum_{i \in I}\left(y_{i,s}|x_{i,s}\right)\rho^2 \log\left(y_{i,s}|\kappa(x_{i,s})\right) \quad (22)$$

$$D'_{x_r \in I}(y_{i,r}|x_{i,r}) = \sum_{i \in I}\left(y_{i,r}|x_{i,r}\right)\rho^2 \log\left(y_{i,r}|\kappa(x_{i,r})\right). \quad (23)$$

When incorporating the GG distribution defined in (22) and (23) for the class-conditional correlation potential of D' in the $i \prec s, i \prec I$ and substituting it into the MeanShift equation (15), the filter for the MRF in the spatial–spectral joint domain can be obtained in $d = p+2$ dimensions, denoted as MS-MRF, as shown in the following equation:

$$M_{h,Q,D'}(x) = \frac{\sum_{i=1}^{n} \boldsymbol{q}_s \boldsymbol{\rho}^2 \log\left(\left\|\frac{\boldsymbol{x}_{i,s}-\boldsymbol{x}_s}{\boldsymbol{h}_s}\right\|^2\right)\boldsymbol{q}_r \boldsymbol{\rho}^2 \log\left(\left\|\frac{\boldsymbol{x}_{i,r}-\boldsymbol{x}_r}{\boldsymbol{h}_r}\right\|^2\right)\boldsymbol{x}_i}{\sum_{i=1}^{n} \boldsymbol{q}_s \boldsymbol{\rho}^2 \log\left(\left\|\frac{\boldsymbol{x}_{i,s}-\boldsymbol{x}_s}{\boldsymbol{h}_s}\right\|^2\right)\boldsymbol{q}_r \boldsymbol{\rho}^2 \log\left(\left\|\frac{\boldsymbol{x}_{i,s}-\boldsymbol{x}_s}{\boldsymbol{h}_s}\right\|^2\right)} - \boldsymbol{x}. \quad (24)$$

The simplified MS-MRF algorithm is shown in Algorithm 2, with the working principle corresponding to Fig. 2. Initially, it determines the resolution level to which the random pixel vector belongs. The rest of the process follows the MeanShift filtering procedure, where the pixel density shifts from low-density areas to high-density areas.

Algorithm 2 MS-MRF

Input

Set parameters:

$\{x_i, x_s, x, v_i, h, q_s, q, Y_j\}$.

$x_i, i = 1, \ldots, n$, is a series of d dimensional independent equally distributed random vectors;

$Y_j, j = 1, \ldots, m$, is a series of d dimensional independent equally distributed random vectors;

$v_i, i = 1, \ldots, n$, is the virtual feature of a series of d dimensional independent identically distributed random x_i;

We define the relationship between space vectors, spectral vector pixels and random vector pixels, and virtual random vector pixels as follows:

$x_i \to v_{i,} v_{i,} \in I$,

$x_s, x_r = \rho^2 \sum i \prec x_i$,

$x_s, x_r = \rho^2 \sum i \prec v_i$.

Output

$$Y_{j+1} = \frac{\sum_{i=1}^{n} q_s \rho^2 \log\left(\left\|\frac{x_{i,s}-x_s}{h_s}\right\|^2\right) q_r \rho^2 \log\left(\left\|\frac{x_{i,s}-x_r}{h_r}\right\|^2\right) x_i}{\sum_{i=1}^{n} q_s \rho^2 \log\left(\left\|\frac{x_{i,s}-x_s}{h_s}\right\|^2\right) q_r \rho^2 \log\left(\left\|\frac{x_{i,s}-x_s}{h_s}\right\|^2\right)} - x$$

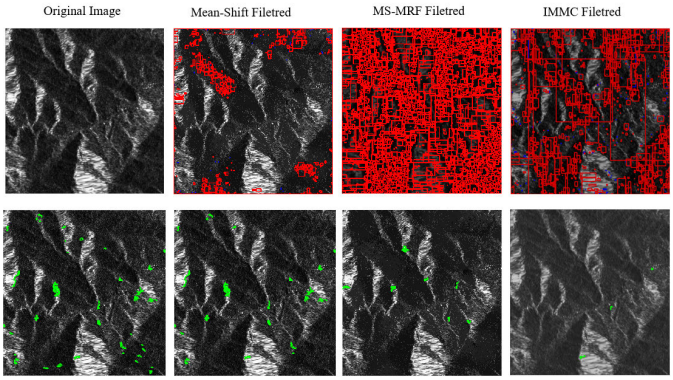


Fig. 3. Interpretability of the IMMC for mountain SAR images.

The MS-MRF filter employs an upward search strategy based on correlated density peaks, which is adaptive and does not have a fixed number of regions. Specifically, the MS-MRF filter dynamically searches for dense regions based on the correlation between pixels. If the correlation between pixels is low, fewer dense regions are searched; conversely, if the correlation is high, more dense regions are searched.

This adaptive search strategy helps reduce noise while strengthening the correlation between point scatters, thereby better preserving edge details. By integrating Algorithms 1 and 2, the MS-MRF method combines the advantages of the MeanShift algorithm and the MRF model. The MeanShift algorithm automatically determines the cluster centers and the number of clusters based on the distribution of pixel values. The MRF model utilizes the correlation between pixels to improve the gray values of adjacent pixels affected by coherence.

To further illustrate, we have provided a specific example showing how the MS-MRF filter dynamically adjusts the search for dense regions under different correlation conditions. For example, Fig. 3 demonstrates the filtering effects in regions

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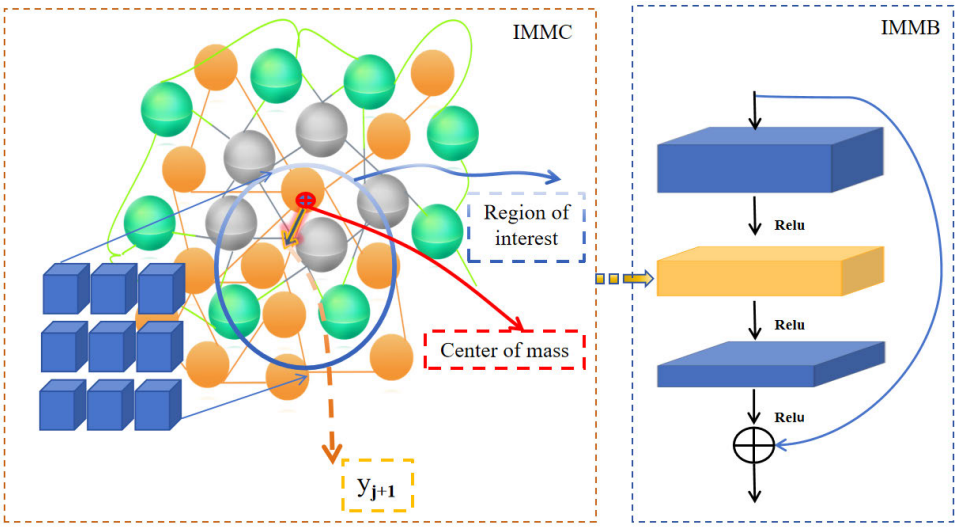


Fig. 4. Interpretive SAR residual filter formation principle.

with high and low correlations, helping readers intuitively understand the advantages of the adaptive search strategy.

D. IMMC

IMMC kernel is realized by integrating the mathematical interpretability of the MS-MRF into a 3×3 convolutional kernel. As depicted in the conceptual diagram of Fig. 4, the mathematical formulation of this process can be succinctly represented as

$$\text{IMMC}(x) = \sum_{k=1}^{K} w_k * \text{MS-MRF}(x). \tag{25}$$

In this equation, w_k denotes the weights of the convolutional kernel, K signifies the total count of kernels employed within the convolutional operation, and MS-MRF(x) represents the outcome of applying the MS-MRF filtering process to the input data x.

First, the input image is divided into several small regions with different densities, the pixels in each region are regarded as a state, and the states of the dense region pixel i and the sparse region pixel S change with the direction of the divided region pixel v. The MS-MRF is constructed by calculating the transfer probability between adjacent pixels in the edge region. This model describes the spatial relationships between pixels.

Second, in terms of the convolution kernel design, the fusion method uses prior knowledge as the weight of the 3×3 convolution kernel. The weight value in the convolution kernel can be adjusted according to the actual situation to achieve different image processing effects.

Finally, the convolution kernel is applied to each pixel with different densities in the image, and the output image is obtained through the convolution operation. During the convolution process, the state transition probability of the current pixel and its surrounding pixels is considered, and the filtering operation is carried out based on the MS-MRF to obtain the feature map with prior knowledge. The advanced nature of the IMMC in the air-optical joint domain is as follows.

1) It can take advantage of the edge-dense Markov model and convolution filter at the same time to extract spatial information in the image and carry out effective image processing. It is more inclined to divide the spatial relationship between the pixels in the dense area to make the edge pixels denser and more significant to better retain the details of the image.
2) The advantage of fusing prior knowledge and the conv3 $\times$ 3 convolution kernel into a convolution filter with the function of prior knowledge is that prior knowledge can be used to extract more accurate image features, while the convolution filter can smooth and sharpen images to further improve image quality.

E. IMMB

The IMMB replaces the 3×3 convolutional kernel within the standard ResNet50 bottleneck layer. The core of this module is the IMMC, which integrates the MS-MRF filtering with convolutional operations. The mathematical expression of the IMMB module can be articulated as follows:

$$\text{IMMB}(x) = \text{ReLU}(\text{IMMC}(x) + x). \tag{26}$$

Here, x represents the input feature map, IMMC(x) denotes the feature map processed by the IMMC, ReLU signifies the rectified linear unit activation function, and the addition symbolizes the residual connection.

To create a portable residual block IMMB oriented toward the residual network structure, compared to directly integrating IMMC into the ResNet network, it can save computational costs and facilitate optimized training for visual interpretation. We replace the original 3×3 convolution in the ResNet bottleneck with IMMC in the spatial–spectral joint domain; after training, the interpretability of the residual filtering IMMB is achieved by visualizing the activation functions of the intermediate layers of ResNet.

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F. Interpretability of ResNet50 (IMMB)

The proposed ResNet50 (IMMB) approach in this article achieves posthoc interpretability by visualizing the activation functions of intermediate layers. This implies that, while the model may not have built-in interpretability during the training process, an explanation of the model's behavior can be provided by analyzing its internal working mechanisms. Posthoc interpretability, facilitated by visual explanation techniques, aims to render the model's interpretability through the visualization of the model's operations. Many visualization methods found in the literature are complemented by dimensionality reduction techniques, which enable simplistic yet human-interpretable visualizations. Visualization may also be employed in conjunction with other techniques to enhance comprehension.

The posthoc interpretability is realized by visualizing the activation maps of the IMMB module, which can be mathematically described through the following steps.

1) *Calculation of Activation Maps:* For each input sample, compute the output activation map of the IMMB module, denoted as $A(x) = IMMB(x)A(x) = IMMB(x)$.
2) *Feature Importance Analysis:* Analyze the activation map A(x) to identify which features are considered significant by the model during the filtering process.
3) *Filtering Effect Evaluation:* Assess the effectiveness of the IMMB module in noise suppression and detailed preservation by comparing the activation maps before and after filtering.

The interpretable filtering process of IMMC on mountain SAR images is shown in Fig. 3 (the red boxes in the figure are different density aggregated regions, the blue dots are the virtual features v_i that delineate the density sparse regions, and the green dots are the presence of noise); the first small figure in the figure is the original image without delineating the density random field; the second small figure is the original MeanShift filtering algorithm pointing to the direction of the highest probability density; the third small figure shows the gradient rising Markov random field search peak method MS-MRF inspired by the variability of substance density, which divides the random fields according to the variability of different substance densities; and the fourth small figure shows the IMMC filtering algorithm, which divides the density random fields in order to differentiate between the different substance edges and denoising while retaining detailed information about the edges.

The ResNet50 (IMMB) architecture enhances interpretability through the incorporation of the IMMB filtering block. This interpretability not only aids in comprehending the operational mechanisms of the model but also bolsters trust in the model's outputs, particularly in applications necessitating a detailed analysis of the processed outcomes. For instance, the IMMB filtering block, by visualizing the activation functions, can demonstrate how the model strikes a balance between smoothing and edge preservation during the filtering process, as illustrated in Figs. 5 and 6. The design of the IMMB filtering block permits the model to maintain high performance while offering profound insights into the filtering procedure,

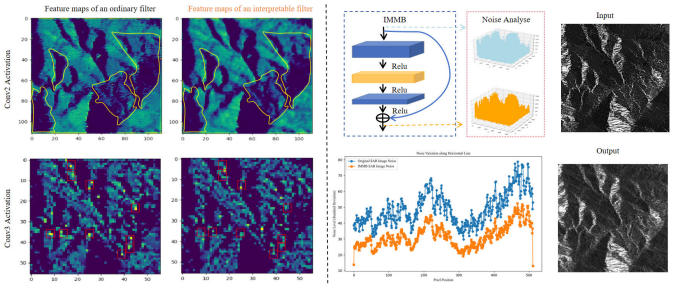


Fig. 5. Interpretable filtering effect of the IMMB on mountain SAR image.

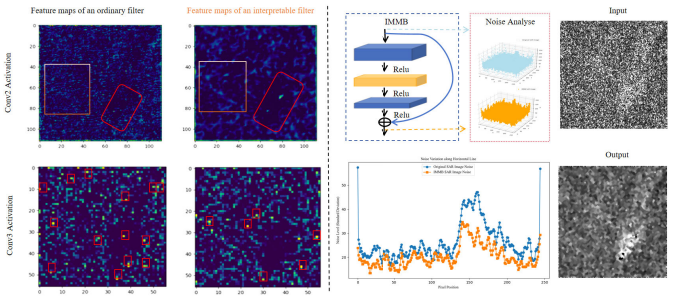


Fig. 6. Interpretability of SAR images of complex coastal ships by the IMMB.

endowing the specific convolutional layers of the ResNet50 (IMMB) with interpretability. This is instrumental in elevating confidence in downstream tasks such as detection and recognition.

Specifically, for ship pixels, a key requirement for reliable SAR filtering is to significantly reduce speckle effects while maintaining resolution and detail. SAR filtering has been extensively discussed and has become relatively mature. The remaining challenge is to effectively and adaptively determine a sufficient number of similar pixels for various textures. As shown in Fig. 5, the interpretable filtering effect of IMMB on SAR mountain maps is illustrated. The activation function feature map conv2 from the visual filter layer IMMC indicates that the filter sharpens edges by delineating edge pixels in neighboring density regions in the same direction while suppressing noise, thereby enhancing edge texture information. Fig. 6 presents the interpretable filtering effect of IMMB on SAR images of complex coastal ships, achieving a good balance between smoothing and edge preservation. The changes in noise contrast values before and after filtering demonstrate the robustness of the IMMC denoising effect.

IV. Experiments

A. Experimental Setup

The experimental environment was configured with CUDA 11.8, PyTorch 2.1.1, and Python 3.10. The relevant experimental parameters for the IMMC were set as follows: $r = 5$, numiter $= 10$, sigma $= 0.5$, bandwidth $= 3$, kernel $= 3$, batchsize $= 16$, lr $= 1e^{-4}$, classes $= 7$, and epoch $= 50$.

To enhance the generalization capability of our proposed ResNet50 (IMMB) model, we have integrated various

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real-world SAR datasets into our research. These include the WHU-SAR-OPT [47] dataset from Wuhan University, which interprets complex terrain features; the AIR-SARShip-1.0 dataset [48] from Chinese Academy of Sciences, focusing on ship detection with images of ships in complex backgrounds; and the OpenSARUrban [57] and OpenSARShip [58] datasets from the Sentinel-1 (S1) satellite, designed for urban interpretation and ship interpretation, respectively, covering a wide range of urban geographical areas. These real SAR datasets provide a more comprehensive testing scenario by covering a broader geographical area, ensuring that our research can be tested under diverse geographical and environmental conditions. To further validate the performance of our model, we have also introduced three synthetic benchmark datasets: Kodak24 [49], McMaster [50], and Set12 [59]. Different noise levels are simulated on these synthetic datasets, with the noise levels for Kodak24 and McMaster set within the range of [1, 10] dB, while the noise level for Set12 is fixed at 30 dB. The key characteristics of the real SAR datasets are summarized in Table I. We randomly selected 300 images from each of the four real SAR datasets, totaling 1200 images. These images were resized into two dimensions: 256×256 and 512×512. Specifically, 200 images from each dataset were resized to 256×256, resulting in 800 images, and 100 images from each dataset were resized to 512×512, resulting in 400 images.

Loss Function: The $L1$-norm loss between the filtered image by ResNet50 (IMMB) and the unfiltered image was used as the loss function L_1

$$L_1 = \frac{1}{N}\|I_O - I_F\|_1 \tag{27}$$

where N denotes the batch size, I_O represents the unfiltered image, and I_F is the filtered image by ResNet50 (IMMB).

After reviewing the literature, we found that most image quality assessment metrics utilize the peak signal-to-noise ratio (PSNR) [51] and the structural similarity index (SSIM) [61]. To facilitate comparison with other works, we selected four metrics to evaluate the quality of real SAR images: PSNR, SSIM, radiometric resolution (RARES) [62], and preservation index (EPI) [63]. PSNR and SSIM are used to assess the quality of synthetic SAR images.

The PSNR is an indicator used to measure image quality by comparing the peak signal power of an image with its mean square error, as shown in (28). The PSNRs of most life scene images are between [20, 40] and [20, 30] dB, indicating that the image quality is average, and values between [30, 40] dB generally indicate that the image quality is good. Below 20 dB, the image quality is poor, and above 40 dB, the image quality is excellent

$$\mathrm{PSNR} = 10 \times \log_{10}\left(\frac{\mathrm{MAX}^2}{\mathrm{MSE}}\right). \tag{28}$$

The SSIM is a metric used to measure the similarity between two images, taking into account not only their brightness but also their contrast and structural information. The SSIM provides a more accurate reflection of changes in image structure and texture. The value of SSIM ranges from $[-1, 1]$, where 1 indicates that the two images are identical.

The RARES is a metric used to measure the gray-level resolution of SAR images. It distinguishes the backscattering coefficients of targets by describing the radiometric quality of each pixel. RARES plays a crucial role in the quantification and interpretation of SAR images. The specific definition is as follows:

$$\mathrm{RARES} = 10 \cdot \log\left(1 + \frac{\sigma}{\mu}\right). \tag{29}$$

The EPI is defined as the ratio of the edges in the filtered image to those in the original image. The closer the EPI value is to 1, the better the edge preservation effect. The specific definition is as follows:

$$\begin{aligned}\mathrm{EPI}_H &= \frac{\sum_i \sum_j \left|\hat{I}(i, j+1) - \hat{I}(i, j)\right|}{\sum_i \sum_j |I(i, j+1) - I(i, j)|} \\ \mathrm{EPI}_V &= \frac{\sum_i \sum_j \left|\hat{I}(i+1, j) - \hat{I}(i, j)\right|}{\sum_i \sum_j |I(i+1, j) - I(i, j)|}.\end{aligned} \tag{30}$$

The EPI_H (horizontal direction EPI) measures the preservation of edge information in the horizontal direction of a processed image. It is calculated by comparing the sum of intensity differences between horizontally adjacent pixels before and after processing. The EPI_V (vertical direction EPI) measures the preservation of edge information in the vertical direction of a processed image. It is calculated by comparing the sum of intensity differences between vertically adjacent pixels before and after processing.

B. Experimental Results and Analysis

To construct an IMMC filter with robust noise suppression and detailed preservation performance, 3×3 convolutions of four different modes (the mean convolution kernel, Gaussian convolution kernel, Sobel convolution kernel, and Prewitt convolution kernel) are renamed C1, C2, C3, and C4, and their corresponding matrices are shown in (31) and (32), respectively, in addition to (33) and (34)

$$\begin{bmatrix} 1/9 & 1/9 & 1/9 \\ 1/9 & 1/9 & 1/9 \\ 1/9 & 1/9 & 1/9. \end{bmatrix} \tag{31}$$

$$\begin{bmatrix} 1/16 & 2/16 & 1/16 \\ 2/16 & 4/16 & 2/16 \\ 1/16 & 2/16 & 1/16 \end{bmatrix} \tag{32}$$

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 5 & -1 \\ 0 & -1 & 0 \end{bmatrix} \tag{33}$$

$$d_x = \begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix} \quad d_y = \begin{bmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}. \tag{34}$$

C1, C2, C3, and C4 are mapped with the MS-MRF to form four kinds of 3×3 mode IMMC filters, namely, IMMC1, IMMC2, IMMC3, and IMMC4. Table II shows the results of the IMMC ablation experiment. The filtering comparison of the four IMMC modes on the original ship image is shown in Fig. 7. Fig. 7 shows a comparison of the noise distributions

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TABLE I
DETAILS OF THE TWO DATASETS

Dataset	Resolution	Size	Sensor	Coverage area	Noise_Level
WHU-SAR-OPT [47]	5 m	5556×3704 px	GF-3	50,000 Km^2	Medium
AIR-SARShip-1.0 [48]	1 m, 3 m	1000×1000 px	GF-3	Multiclass sea state	Serious
OpenSARUrban [57]	Multiple resolutions	Size of change	S1	Urban area	Change
OpenSARShip [58]	Multiple resolutions	Size of change	S1	A sea area containing ships	Change

TABLE II
ABLATION COMPARISON OF THE FOUR IMMC FILTERS

Methods	PSNR ↑	SSIM(%) ↑
IMMC1	32.24	62.90
IMMC2	**34.63**	**66.57**
IMMC3	21.99	46.34
IMMC4	20.10	35.01

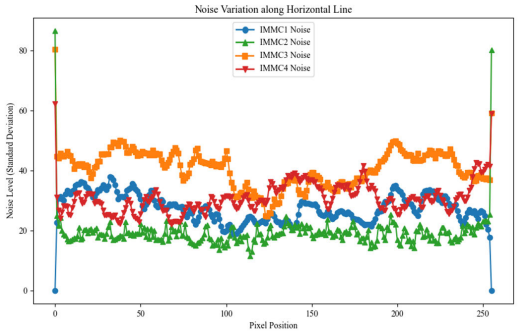


Fig. 7. Comparison of the noise distributions in the four IMMC filtered images.

TABLE III
ABLATION EXPERIMENT

Methods	PSNR ↑	SSIM(%) ↑
MS-MRF	30.74	46.34
IMMC	**34.63**	**66.57**
ResNet50	30.57	57.36
ResNet50 (IMMB)	**36.21**	**68.95**

of four of the IMMC-filtered images, corresponding to Noise image1, Noise image2, Noise image3, and Noise image4.

By comparing and analyzing the four IMMC-filtered image quality evaluation objective index values in Table II, we find that the five indices of IMMC2 with the Gaussian convolution kernel are significantly better than those of the other three. By directly comparing the effects of the four filtered IMMC images in Fig. 7 with human eyes, we find that the filtered IMMC2 images with a Gaussian convolution kernel as the carrier kernel are more consistent with the recognition of environmental and target separation by human eyes. By statistical comparison of the noise distribution map after IMMC filtering.

Table III presents the ablation study results for the proposed interpretable filtering method. An objective analysis of the quality assessments indicates that the proposed method shows enhanced capabilities in noise suppression and detail preservation. To further validate the robustness of the proposed method, it will be compared with numerous traditional methods and state-of-the-art (SOTA) approaches.

TABLE IV
COMPARATIVE EXPERIMENTAL DIAGRAM OF HETEROGENEOUS REGIONS

Methods	PSNR ↑	SSIM(%) ↑	RaRes	EPI (%)
Lee [5]	20.71	62.90	1.75	86.88
Frost [6]	21.03	61.35	2.00	92.74
Kuan [7]	22.30	36.24	**2.32**	97.84
Bilateral	21.44	41.29	2.03	95.95
MMSE [8]	19.30	40.12	2.06	94.57
NL-SAR [55]	24.73	83.69	1.83	96.10
MS-MRF	**25.57**	**84.35**	2.16	**98.12**

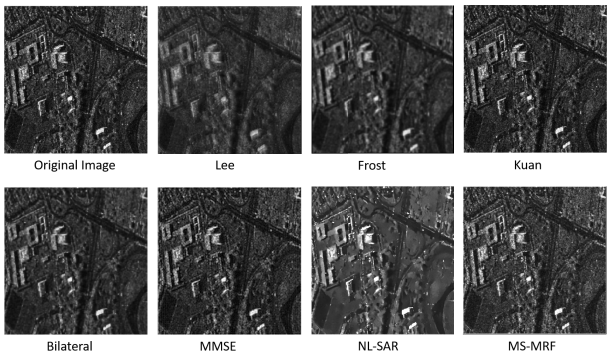


Fig. 8. Experimental comparison of TSX slice containing heterogeneous regions with classical filters.

To objectively demonstrate the robustness of the MS-MRF filter, the proposed MS-MRF filter was compared with six classical filters (Lee, Frost, Kuan, Bilateral, MMSE, and NL-SAR) under noisy conditions. This comparison was conducted to verify the robustness of the proposed filter in heterogeneous regions. The images used for testing were sourced from the TerraSAR-X (TSX) dataset [56]. The comparative chart after filtering by the seven filters is presented in Fig. 8. Upon visual inspection and comparison with the filtering quality of the six traditional spatial filters, it was observed that the images filtered by MS-MRF exhibit clear edge textures and retain detailed information. Furthermore, the image quality postfiltering was objectively assessed using evaluation metrics, with the results displayed in Table IV. Upon comparison, it was determined that the MS-MRF filter yielded the best results. The analysis, both subjective and objective, based on the aforementioned charts, leads to the conclusion that the MS-MRF filter possesses robust filtering capabilities in heterogeneous regions.

Due to coherence, the gray values of adjacent pixels exhibit random variations, which are centered around a mean value, resulting in speckle noise in images. Traditional SAR image filtering techniques often overlook the coherent nature of

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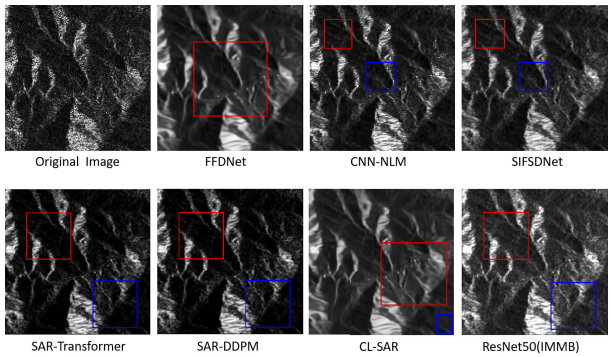


Fig. 9. Filtering performance of SOTA SAR denoising models on mountainous SAR images (512 size) from the WHU-SAR-OPT dataset.

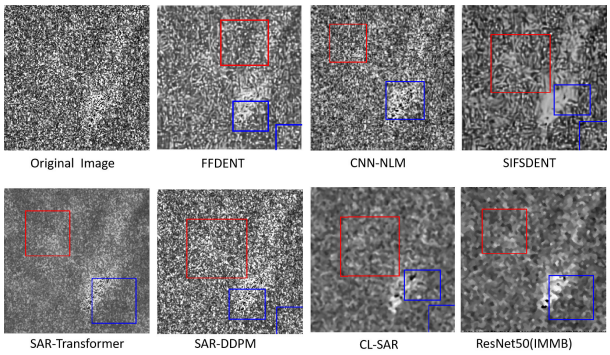


Fig. 10. Filtering performance of SOTA models on the AIR-SAR-Ship dataset.

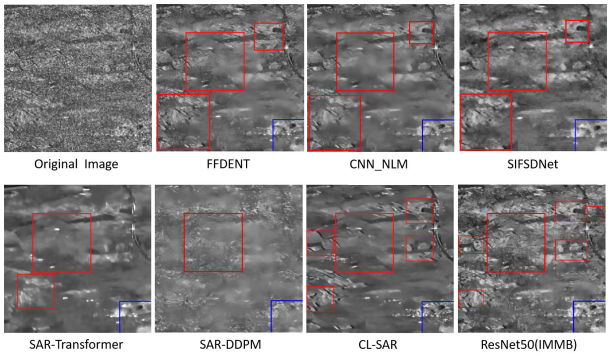


Fig. 11. Filtering performance of SOTA models on the OpenSARUrban.

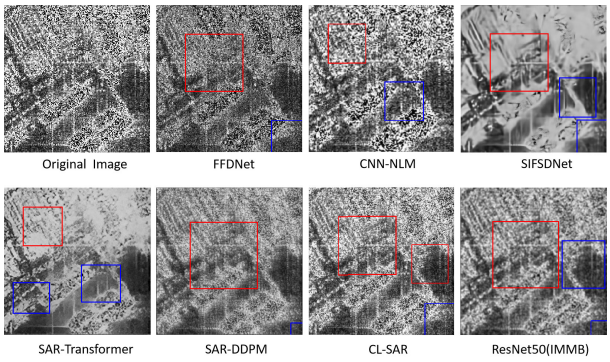


Fig. 12. Filtering performance of SOTA models on the OpenSARShip.

noise, leading to the loss of important details during the filtering process. By comparing with traditional filters, we found that the MS-MRF denoising filter combines the advantages of the MeanShift algorithm and the MRF model. The MeanShift algorithm automatically determines the cluster centers and the number of clusters based on the distribution of pixel values, while the MRF model utilizes the correlation to improve the gray values of adjacent pixels affected by coherence. This combination, compared to other traditional filters, effectively removes speckle noise in SAR images while preserving edge and texture features.

To objectively verify the advanced performance of our proposed ResNet50 (IMMB) filter, we compared it with six DL filtering techniques (FFDNET [52], CNN-NLM [53], SIFSDNET [24], SAR-Transformer [65], SAR-DDPM [54], and CL-SAR [27]) across four real datasets. The filtering results are shown in Figs. 9–12. Visual analysis indicates that under moderate to severe noise conditions in SAR real scenes, the image filtered by our ResNet50 (IMMB) demonstrates superior quality compared to the other six DL filtering techniques. This enhanced image quality is beneficial for improving the performance of downstream decision-making tasks. Objective analysis of the PSNR, SSIM, RARES, and EPI evaluation results further confirms the robustness of the ResNet50 (IMMB) filtering model in terms of noise peak suppression, structural preservation, scattering characteristics, and edge retention. As shown in Table V, the comparison results of SAR denoising models on different slices of the four real datasets are presented.

When performing noise filtering, an important objective is to preserve the original scattering characteristics of the SAR image after filtering. This means that the filtering algorithm must suppress noise while retaining the original properties of the scatter in the image, such as intensity and positional information [63]. Unlike optical images, the physical meaning of pixel values in SAR images differs from that in natural scene images due to the diversity of target scattering mechanisms [64].

In SAR imagery, the intensity and positional information of point scatters are crucial for image interpretation. The four real SAR datasets exhibit varying noise levels, such as samples of mountains from WHU-SAR-OPT, samples of ships on the sea from AIR-SARShip-1.0, urban areas with buildings and vegetation from OpenSARUrban, and samples of ships docked by the shore from OpenSARShip, as well as samples of a house from the synthetic dataset Set12. Through visual observation and analysis, the following conclusions were drawn: the ResNet50 (IMMB) demonstrates robust filtering performance on SAR images composed of a single material and a homogeneous background (samples of mountains from WHU-SAR-OPT, ships on the sea from AIR-SARShip-1.0, and houses from Set12): it reduces the coherent speckle noise in the background areas and enhances the scattering correlation at the edges of object areas, thereby improving edge preservation performance.

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TABLE V
COMPARISON OF SAR DENOISING MODELS ON THE REAL SAR DATASET RESULTS (PSNR ↑| SSIM(%) ↑) (RARES ↑ _ EPI(%) ↑)

Datasets	FFDNET [52] (2019)	CNN-NLM [53] (2020)	SIFSDNET [24] (2022)	SAR-Transformer [65] (2022)	SAR-DDPM [54] (2023)	CL-SAR [26] (2024)	**ResNet50 (IMMB)** (Ours)
WHU-SAR-OPT	14.94|66.28	14.94|57.44	14.09|41.39	13.02|52.18	12.91|40.81	14.93|57.70	**22.90|69.77**
	2.75-98.30	2.79-96.65	2.40-97.37	1.40-97.37	2.63-98.25	2.54-90.39	**3.84-98.56**
AIR-SARShip-1.0	12.61|70.30	14.20|37.08	10.20|30.84	14.07|41.39	12.91|40.81	14.92|57.77	**15.72|62.21**
	1.57-88.77	1.64-96.42	0.99-89.41	1.69-95.91	1.00-82.78	1.76-81.50	**1.78-98.97**
OpenSARUrban	22.22|55.33	16.73|41.35	18.75|13.30	18.18|27.04	19.34|20.97	18.62|16.30	**21.64|68.95**
	0.71-91.89	0.53-89.00	0.58-85.81	0.65-87.84	0.52-81.91	0.38-78.87	**0.86-94.66**
OpenSARShip	14.82|43.02	14.27|57.48	15.94|71.30	14.93|66.28	12.67|70.02	16.66|80.36	**18.12|83.68**
	1.31-97.74	1.56-94.94	1.15-93.17	1.40-96.14	1.35-97.47	1.33-98.80	**1.61-99.33**

TABLE VI
COMPARISON OF SAR DENOISING MODELS ON SYNTHETIC DATASETS UNDER DIFFERENT NOISE LEVELS (PSNR ↑| SSIM(%) ↑) RESULTS

Datasets	Looks	FFDNET	CNN-NLM	SIFSDNET	SAR-Transformer	SAR- DDPM	CL-SAR	**ResNet50 (IMMB)**
McMaster	L = 1	31.04|81.33	29.80|86.70	33.22|82.18	33.52|71.51	34.43|72.10	34.73|83.54	**35.09|87.07**
	L = 5	31.26|81.96	30.51|87.28	33.54|82.18	34.74|72.75	35.59|83.88	35.71|83.98	**35.74|87.37**
	L = 10	31.87|82.72	31.36|88.03	33.78|82.18	33.74|73.04	36.13|74.20	36.27|85.03	**36.39|87.84**
Kodak24	L = 1	32.42|81.47	34.60|82.49	35.36|84.93	33.17|72.31	33.38|78.34	35.64|86.37	**36.52|88.21**
	L = 5	33.75|84.60	35.27|83.61	36.08|86.35	35.87|76.39	35.75|85.88	36.29|87.09	**37.10|89.25**
	L = 10	34.87|85.39	35.85|86.39	37.28|87.67	37.79|77.14	37.08|88.18	37.21|88.48	**38.29|91.68**

TABLE VII
COMPARISON OF SOTA SAR DENOISING MODELS ON SET12 "HOUSE" IMAGE AT 30-DB NOISE LEVEL

Datasets	FFDNET	CNN-NLM	SIFSDNET	SAR-Transformer	SAR- DDPM	CL-SAR	**ResNet50 (IMMB)**
Set12	14.09|26.22	14.36|56.78	14.76|36.79	13.71|24.21	13.05|7.48	**16.03**|45.35	14.9|**69.74**

Specifically, for ship pixels, the majority of pixels originating from the ship target [68] are identified as similar samples, while any pixel from the ocean background is excluded. Conversely, for ocean pixels, most of the ocean pixels surrounding the ship target are deemed similar samples, with all ship pixels and edge pixels excluded. Additionally, for edge pixels, only a portion of the edge pixels between the ship and the ocean background are gathered as similar samples.

The ResNet50 (IMMB) also outperforms other SOTA filtering methods on SAR images composed of two or more heterogeneous objects. Point scattering in heterogeneous areas (such as OpenSARUranban) filters differently in different light and dark areas [68], and the filter effect is more obvious in brighter areas, because different material pixels in dark areas are strongly correlated and dense, and there is an undeniable challenge in searching upward for dense centers of correlation.

Tables V–VII demonstrate that the ResNet50 (IMMB) outperforms or is at least comparable to the current best methods in terms of PSNR and SSIM metrics across multiple real SAR datasets (such as whu-sar-opt, AIR-SARShip-1.0, OpenSARUrban, and OpenSARShip). Moreover, as shown in Table V, the RARES and EPI metrics of ResNet50 (IMMB) are superior to the other six filtering methods. This indicates that ResNet50 (IMMB) possesses robust performance in noise suppression and detail preservation.

The other DL approaches mentioned in Tables V–VII, such as FFDNET, CNN-NLM, SIFSDNET, SAR-Transformer, SAR-DDPM, and CL-SAR, are often considered as "black box" models. While these models may exhibit excellent performance, they lack an intuitive mathematical interpretation and a transparent working process, which limits the understanding of the model's internal mechanisms. For instance, models like SAR-Transformer and SAR-DDPM may perform well on specific tasks but lack an intuitive explanation of the filtering process.

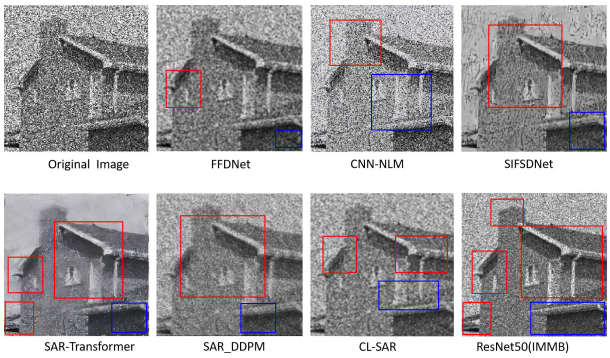


Fig. 13. Demonstration of the filtering effect of SOTA SAR denoising models on Set12 "House" image at 30-dB noise level.

By comparing the filtering effects from Figs. 9 to 13, it is evident that the ResNet50 (IMMB) exhibits exceptional performance in noise suppression. Compared to the original images, the images filtered by the IMMB module demonstrate clearer structural features and a significant reduction in random speckles, indicating that the IMMB module effectively lowers the noise level in the images. In contrast to other SOTA models, the IMMB maintains image clarity while better preserving image details.

In SAR imagery, the intensity and positional information of point scattering are crucial for image interpretation.

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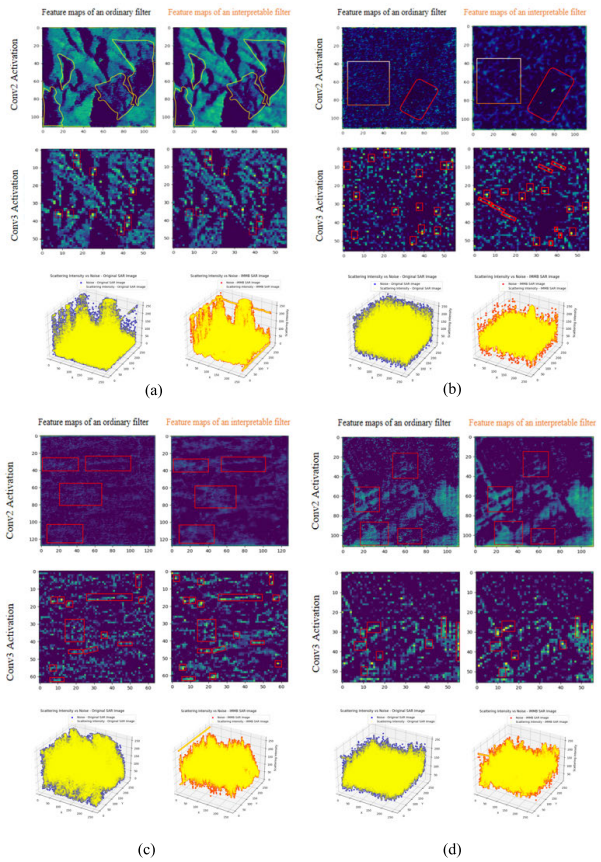


Fig. 14. Posthoc interpretability of the IMMB for four real SAR samples. (a) WHU-SAR-OPT, (b) AIR-SAR-SHIP, (c) S1-OpenSARUrban, and (d) S1-OpenSARShip, to ensure that different colors are used to represent noise and scattering intensity; specifically, we will use blue to indicate the noise in the original SAR images, yellow to represent the scattering intensity, and orange to signify the noise in the IMMB SAR images.

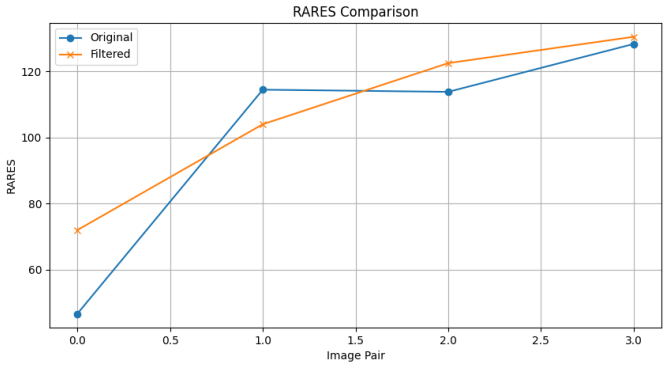


Fig. 15. ResNet (IMMB) demonstrates its scattering statistics before and after filtering on four real SAR images.

Figs. 9–13 highlight the performance of different models in edge and detailed preservation with red and blue boxes. The ResNet50 (IMMB) shows in these critical areas that the model not only filters out noise but also effectively retains the original characteristics of point scatters, which is extremely important for subsequent image analysis and interpretation tasks.

In Fig. 14, the filtering effect of the ResNet50 (IMMB) on four real SAR samples can be observed: In the Conv3 activation maps, the image after filtering by the IMMB (orange) exhibits a significant reduction in the number of noise points compared to the original image (blue), indicating a marked decrease in noise levels.

The ResNet (IMMB) demonstrates its scattering statistics before and after filtering on four real SAR images, as shown in Fig. 15. It enhances the suppression of noise to a certain extent while preserving the scattering characteristics of the image, particularly in the brighter regions. This is due to the DL model's ability to learn which features are important and retain them during the processing.

V. Conclusion

In this article, we primarily proposed a mathematically interpretable MS-MRF SAR image filter. Building upon the MS-MRF, we conducted in-depth research and constructed the IMMC filter and IMMB filtering block in the spatial–spectral joint domain. Transplanting them into DL models yields an explainable filtering model. The IMMC is a 3×3 SAR denoising convolutional kernel. This new convolutional kernel is designed based on the unique physical imaging characteristics of SAR, combining a statistical filter that searches for the highest density correlation with convolution, offering greater generalization compared to traditional SAR denoising algorithms, and achieving ideal filtering results in both homogeneous and heterogeneous regions.

By comparing with classical filtering and advanced DL filtering techniques, it was found that the ResNet50 containing the IMMB filtering block achieved robust results on both real and synthetic datasets. A series of large-scale experiments shows that, compared with the SOTA SAR denoising methods, the average SSIM of the proposed method is more than 85.00% and the average PSNR is more than 35.00 dB on synthetic datasets with different noise levels, and it has significant performance advantages in noise suppression, detailed preservation, and interpretability. The filtered images are more in line with the imaging mechanism of SAR images, and the texture structure is more consistent with human visual perception, which is of great significance for downstream tasks such as change detection and target detection.

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